Talk Like a Structured Graph: Augmenting Text Encodings with Structural Statistics

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TL;DR. *We investigate whether prepending graph features — node degree, local clustering coefficient, and random-walk return probability — to graph text encodings improves LLM accuracy on GraphQA-style reasoning tasks, and whether this benefit depends on task alignment and model capacity.*

1 Background & Motivation

Large language models applied to graph-structured data through text encodings perform poorly on even basic structural reasoning tasks. Fatemi et al. [2023] show that black-box LLMs underperform a simple majority-class baseline on tasks like edge existence and cycle check, and that the choice of graph encoding function alone can shift accuracy by tens of percentage points on the same underlying task. This leaves an open question: is the bottleneck the encoding style (how the graph is phrased), or the information content available to the model at inference time?

This question matters because black-box LLM access is now the dominant way most practitioners interact with these models. Any technique that improves graph reasoning without fine-tuning or architectural access is immediately useful wherever graph-structured data needs to be handed to an off-the-shelf LLM, from knowledge-graph question answering to social-network analysis pipelines.

Prior work leaves this gap open in a specific way. Wang et al. [2023] established that basic graph reasoning remains difficult for LLMs, motivating benchmarks like GraphQA. Fatemi et al. [2023] build on this by varying how a fixed graph is phrased as text, but never vary what structural information is present beyond the raw edge list.

Separately, Rampášek et al. [2022] show that explicit structural encodings — most notably random-walk structural encoding (RWSE) — meaningfully improve a graph transformer's performance, but only as learned numerical embeddings internal to a trained model, never as information exposed to a black-box text interface. No existing work tests whether the same structural signal, expressed instead as a natural-language statement and prepended to a text encoding, transfers any benefit to an LLM that cannot be fine-tuned or given internal access. That is the gap this project addresses: concretely, we build directly on the framework of Fatemi et al. [2023] and aim to improve on their reported accuracy for the same GraphQA tasks and encoding functions, using structural primers as the added ingredient.

2 Proposed Approach

- **Core idea** — We treat structural graph statistics as a natural-language "primer": a short preamble of factual sentences about each node's local structure, prepended before the standard graph text encoding and the task question. We measure whether this primer improves LLM accuracy on GraphQA-style tasks relative to the same encoding with no primer, and whether the effect depends on how closely the primer's content matches the task being asked.
- **Hypotheses** — We predict primer benefit tracks task alignment: largest on primer-adjacent tasks (connected nodes, edge count), where the primer supplies usable ingredients but the model must still combine them; near-zero on primer-agnostic tasks (node count, cycle check, edge existence); and trivially high on the aligned task, serving only as a manipulation check. A benefit on the agnostic tasks would be the

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most surprising and informative outcome. We further expect the benefit to shrink with model capacity, as larger models recover more structure unaided.

- **Technical outline** —
  - **Base encoding:** Incident encoding from Fatemi et al. [2023], fixed across all conditions.
  - **Primer conditions:** no primer (control); degree only; clustering coefficient only; random-walk structural encoding (RWSE) only; all three combined. Each primer is one short factual sentence per node. Degree and clustering are a single number per node; RWSE [Rampášek et al. 2022] is the per-node vector of return probabilities at walk lengths $k = 1, \ldots, K$ ($K = 4$), listed at two-decimal precision.
  - **Tasks:** all six basic GraphQA tasks — node degree, connected nodes, edge count, node count, cycle check, and edge existence — spanning the primer-alignment spectrum, from primer-aligned (node degree) through primer-adjacent (connected nodes, and edge count via the sum of degrees) to primer-agnostic (node count, cycle check, edge existence). The aligned task, where the degree primer states the answer almost verbatim, serves as a manipulation check that the model reads and uses the primer at all; the scientifically informative test lies in the adjacent and agnostic tasks, where a gain cannot be explained by the primer trivially containing the answer.
  - **Graphs:** the pre-generated graphs from the published GraphQA dataset of Fatemi et al. [2023] (released on Hugging Face), reused unchanged across all primer conditions — so the graph-structure effect they document cancels in the paired with-versus-without-primer comparison.
  - **Models:** two open model families at two sizes each — Gemma 4 (4B and 12B) and Qwen3 (8B and 14B) — giving a within-family capacity comparison in both families and a cross-family check, all queried at temperature 0, no fine-tuning.
  - **Prompting:** each condition is run under both zero-shot and chain-of-thought (CoT) [Wei et al. 2022] prompting. We cross these because a primer can change what reasoning a task requires: exposing per-node degrees, for instance, turns edge count into a multi-step arithmetic sum (degrees over two), the multi-hop regime where CoT is expected to help. Zero-shot alone would conflate "the primer does not help" with "the model had no room to use it," so we measure both.
- **Novelty** — This design connects the two lines of work directly: it takes a structural encoding (RWSE) originally designed to improve a trained graph transformer's internal representations [Rampášek et al. 2022] and repurposes it as a natural-language primer for a black-box LLM prompt (in the spirit of Fatemi et al. [2023]) — without any model training. To our knowledge, no existing work has tested whether this specific class of structural signal transfers when moved from a learned embedding to explicit text.

3 Experimental Plan

- **Datasets** — We use the published GraphQA dataset of Fatemi et al. [2023] on Hugging Face, which ships pre-generated graphs (Erdős–Rényi, Barabási–Albert, and others) already paired with per-task questions and ground-truth answers. From its held-out test split we sample 30 examples per task across the six tasks (180 graph–task instances), a starting budget we will increase if compute allows; we generate no graphs, only computing each primer's statistics from the provided edge lists with NetworkX [Hagberg et al. 2008]. Each instance is run under all five primer conditions in a paired design.
- **Baselines / comparisons** — The primary comparison is internal and paired: each primer condition is measured against an otherwise-identical query on the same graph with the structural primer removed (the no-primer Incident encoding), so any accuracy difference is attributable to the added structural information alone — e.g., the random-walk-only condition against the same query without it. We additionally report a majority-class baseline computed over the sampled examples for cycle check and edge existence, since

both tasks have a known class-imbalance pitfall [Fatemi et al. 2023]. Where relevant, we note the original paper's published PaLM-based numbers as a rough external reference point, explicitly flagged as not a strict comparison given the difference in model scale.

- **Metrics** — Node degree, node count, and edge count are scored by exact-match accuracy (with mean absolute error as a secondary signal); connected nodes by set-level F1 against ground truth; cycle check and edge existence by accuracy reported alongside the majority-class baseline. Because each graph is evaluated under all five primer conditions, we test each condition against the no-primer control per (task, model, prompting style) using McNemar's test, appropriate for this paired binary-outcome setup.
- **Models** — Two open model families at two sizes each: Gemma 4 (4B and 12B) and Qwen3 (8B and 14B). This lets us test within each family whether primer benefit shrinks as model capacity grows, while comparing across families guards against family-specific effects. Every (primer, task, model) cell is run under both zero-shot and chain-of-thought prompting, letting us separate whether a primer helps in a single forward pass from whether it helps once the model is given room to reason. All models are queried with temperature fixed to 0.

References

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